



Brain-machine link

In a move to lubricate the integration of robotics to human body Australian scientists have developed an interface device called 'Stentrode'. <http://dgit.in/Stentrode>



Wet Data

After sky, water is the next step to connect you better is what Microsoft is thinking with its underwater data centers. <http://dgit.in/DataWet>

ing their satellites closer to home, both SpaceX and OneWeb could bring down latency from 500 milliseconds to 20 milliseconds – roughly the time equivalent for a home internet connection that works with fiber optic in the US.

But before you think that this is the solution that puts satellite internet back in business, there's something else you ought to know: the signal from a satellite in low Earth orbit won't cover as much surface area of the Earth as a satellite in geosynchronous orbit (about 22,000 miles above Earth). This means more satellites need to be deployed. Even though the price of satellites have come down in the last two decades, it still may not be cheap enough to allow for a significantly larger number of satellites being deployed in such a short timeframe.

Augmenting the skeptic's view is also the matter of failed attempts in the past. For this isn't the first time that the attempt to bring high speed internet with the aid of satellites is being made. Multiple companies tried to deploy satellites into low Earth orbit with the same intention in the 1990s. One of these companies was Teledesic which was funded by Bill Gates among others. The company planned the launch of 840 satellites but faced setbacks only to cancel its operations in 2002 followed by relinquishing their wireless spectrum rights to Federal Communications Commission (FCC) the very next year.

And then we come to the money part

As mentioned before, there's the cost incurred with so many satellites (fittingly enough, the figure will be astronomical) that's a cause for worry for the companies involved. Yet another problem area that the firms will face is with controlling the



Money - always a key player



SpaceX-charting the heavens

costs of the internet which they're going to provide for the billions who don't have internet access at the moment. And these billions are the poor in developing countries for whom subscription, satellite dishes etc. must come at an affordable price.

Also, fiber optic internet connections and wireless mobile data connections are gaining traction throughout the world. This points to the possibility that these modes of connecting the world would prove to be cheaper than serving them using satellites.

The shape of things to come

By this time, it has become rather evident that the space race to bring internet to a larger population is rife with challenges. But the fascinating thing is that companies are still not backing down from the endeavour. Many believe that they will only realise that the costs involved are too high when it's already too late.

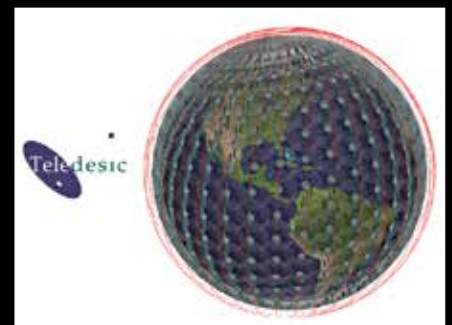
They aren't even backing down after knowing the technical hurdles that are involved, which makes us wonder what it is that they know that we don't.

Hopeful as the fruition of the process seems to be, one aspect that should be taken particular note of is that in this version of the space race – unlike the one between America and Russia that took place during the 1950s – it's billion dollar companies that are the key players. Governments have been reduced to being mere facilitators.

It's rather obvious to most that the future of the world will be determined by how much we explore (or exploit) resources that are outside our planet. The companies that get a first-mover advantage in the space

realm will have a huge impact in shaping the world to come. Should the governments sit back and not have a stake in this?

It's not just in terms of satellite internet that private firms have become the main players. Even the arena of rockets, which have traditionally been the domain of nation-states, is experiencing this paradigm shift. Recently, Blue Origin a firm established by Amazon.com founder Jeff Bezos was able to successfully launch a rocket to suborbital space and brought it back down again for a soft landing – making it the world's first reusable rocket. This is something that could make space exploration much more economically feasible. This is the kind of technology one would expect a government agency such as NASA to come up with, rather than a private firm – especially since the government agencies have been working in the field for a much longer time, and also because they have been complaining about their budgets being cut by governments for the longest time.



The failed plan of Teledesic

Whatever the case, a clear picture emerges that paints mankind's future in space as being directed by firms driven by the eventual goal of making a profit. This raises some interesting social questions, such as, is it wise to let the interests of corporations dictate the space race, or should we focus more on the needs of the common people?

It is, after all, the mistakes made by governments and the greed of corporations and individuals that's caused all of the inequality and suffering of the poor. Is it really wise to expect a solution to emerge by using the same methods that caused the problem in the first place? Only time will tell... 